

Topologically nontrivial sectors in the Abelian Higgs model with massless fermions

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(Received 31 May 1991)

We develop an approach to the dilute-instanton-gas approximation for the Abelian Higgs model when massless fermions are present. We discuss the properties of the resulting partition function and evaluate fermionic correlation functions which become nontrivial when topological sectors are taken into account.

I. INTRODUCTION

A path-integral approach to the study of two-dimensional fermionic models in nontrivial topological sectors has been recently developed in Refs. [1–3]. Basically, the method starts from the observation that if one writes the vector field carrying the topological charge [a gauge field in the case of the Schwinger model, two-dimensional QCD (QCD₂), etc., an auxiliary field in the case of the Thirring model, the chiral Gross-Neveu model, etc.] in the form

$$A_\mu = A_\mu^{\text{cl}} + a_\mu, \quad (1.1)$$

with A_μ^{cl} a fixed (background) configuration carrying the topological charge of A_μ , then a_μ can be completely decoupled from fermions and the models can be analyzed in a very simple way. Indeed, in two-dimensional space-time a_μ can be written in the form

$$a_\mu = e \gamma^\mu a_\mu = e^{-i\eta} e^{\gamma_5 \varphi} i \not{\partial} e^{\gamma_5 \varphi} e^{i\eta} - e^{-i\eta} i \not{\partial} e^{i\eta}, \quad (1.2)$$

where (η, φ) takes values in the algebra of the gauge group and e is the gauge coupling constant. Note that since A_μ^{cl} carries the whole topological charge, a_μ lives in the zeroth topological sector and hence φ can be globally defined. Once a_μ is written in the form (1.2), it can be decoupled from fermions $\psi, \bar{\psi}$ just by performing the change of variables:

$$\psi \rightarrow e^{-i\eta} e^{-\gamma_5 \varphi} \psi, \quad (1.3a)$$

$$\bar{\psi} \rightarrow \bar{\psi} e^{-\gamma_5 \varphi} e^{i\eta}, \quad (1.3b)$$

As usual, quantum aspects of this change of variables are taken into account through the associated Fujikawa's Jacobian [4]. Had we written the complete gauge field A_μ in a form analogous to (1.2), then the corresponding φ would have implemented a topology-changing transformation, and in that case the evaluation of fermionic Jacobians would have become more involved [3].

Using this approach, various fermionic models have been investigated and an interesting structure arising from topological sectors has been studied in a very simple and transparent way. In particular it was shown that certain fermionic correlation functions for the Schwinger and QCD₂ models [1–3] become nontrivial when sectors

with topological charge $n \neq 0$ (i.e., instanton contributions) are appropriately taken into account.

It is the purpose of this work to discuss these issues in the two-dimensional Abelian Higgs model coupled to massless fermions. In particular, we shall study the contribution of topological sectors to the partition function as well as certain fermionic correlation functions which become nontrivial only when instanton contributions are appropriately taken into account.

The Abelian Higgs model in two dimensions has attracted much attention since Nielsen and Olesen [5] discovered vortexlike solutions to its classical equations of motion. These solutions, carrying a topological charge (the number of magnetic flux units of the vortex configuration), can be used as instanton in the analysis of the (Euclidean) path-integral defining the partition function. In this way it was shown within a dilute-instanton-gas approximation [6–9] that the vortex contribution restores the originally broken U(1) symmetry of the Abelian Higgs model. Consequently, fractional external electric charges are confined by a linear potential. In Refs. [7] and [8] the analysis was extended to include massless fermions minimally coupled to the U(1) gauge field. Statistical mechanical analogies (basically the Coulomb gas and the X-Y models) indicate that, in this case, certain instanton effects are wiped out [7,8].

The model has regained interest in recent days in connection with the possibility of fermion-number nonconservation at high temperatures or high energies in the standard electroweak theory [10–14], due to instanton effects. Since no instanton solution exists in the standard model, *constrained* instantons have to be introduced and the corresponding calculation is complicated. This problem does not arise in the two-dimensional Higgs model for which even an exact solution is known, when a certain relation between the gauge-boson mass and the Higgs-boson mass holds [15]. Again, the model has been used in Refs. [16] and [17] as a simplified laboratory to study whether fermion-number nonconservation, in principle suppressed by an exponentially small WKB factor [18], is enhanced at high energies. Different approximations have been used in these works for evaluating instanton effects, but they are restricted to the one-instanton sector. In this sense, it is important to investigate how a dilute-gas approximation (in which one considers a superposition of instantons and anti-instantons) affects the re-

sults; in particular, whether the resulting nonlocality of the relevant vertex, induced by different (anti-)instantons saves unitarity at high energies (the leading-order result exceeds the unitarity bound [16,17]).

In the present work we develop a systematic approach to the dilute-gas approximation for the two-dimensional Abelian Higgs model with massless fermions. After a discussion of the decoupling of fermions from gauge fields in Sec. II, we analyze in Sec. III the partition function for the model in the dilute-gas approximation. As we discuss at the end of the section, our calculations confirm the scenario resulting from statistical mechanical analogies [7,8]. In fact, our approximation yields to a partition function which can be related to the Coulomb gas or X - Y model partition function, although it corresponds to a more complicated Hamiltonian. However, since the logarithmic interaction dominates this Hamiltonian at large distances, the results inferred from Coulomb gas and X - Y models also hold in our approximation. We then proceed in Sec. IV to the evaluation of fermionic correlation functions which become nontrivial when contributions from topological sectors are taken into account.

II. THE PARTITION FUNCTION

Let us consider the partition function for the two dimensional (Euclidean) Abelian Higgs model coupled to massless Dirac fermions:

$$Z = \int D A_\mu D \phi D \bar{\psi} D \psi \exp \left[- \int d^2 x (L_{\text{Higgs}} + \bar{\psi} \not{D} [A] \psi) \right], \quad (2.1)$$

where

$$L_{\text{Higgs}} = -\frac{1}{4} F_{\mu\nu}^2 + \frac{1}{2} |D_\mu \phi|^2 - c_2 |\phi|^2 + c_4 |\phi|^4 + \frac{c_2^2}{4c_4} \quad (2.2)$$

and

$$D_\mu = i \partial_\mu + e A_\mu. \quad (2.3)$$

In (2.1) fermions have a *flavor* index, $\psi = \psi^a$, with $a = 1, \dots, N_F$.

Let us write the contribution to Z coming from different topological sectors in the form

$$Z = \sum_n \int D A_\mu^{(n)} D \phi^{(n)} D \bar{\psi} D \psi \exp(-S[A_\mu^{(n)}, \phi^{(n)}, \bar{\psi}, \psi]), \quad (2.4)$$

where

$$S[A_\mu^{(n)}, \phi^{(n)}, \bar{\psi}, \psi] = \int d^2 x (L_{\text{Higgs}}[A_\mu^{(n)}, \phi^{(n)}] + \bar{\psi} \not{D}[A^{(n)}] \psi), \quad (2.5)$$

and $(A_\mu^{(n)}, \phi^{(n)})$ represents a gauge-Higgs-field configuration belonging to the n th topological sector; i.e.,

$$\oint_C A_\mu^{(n)} dx^\mu = -\frac{2\pi n}{e}, \quad (2.6)$$

with C a circle at infinity ($C = S_\infty^1$) and the scalar field configuration satisfying

$$D_\mu [A_\mu^{(n)}] \phi^{(n)}|_{S_\infty^1} = 0. \quad (2.7)$$

The asymptotic form of such a configuration, a vortex line, is

$$A_\mu^{(n)} = \frac{n}{e} \epsilon_{\mu\nu} \partial_\nu \alpha_{\text{cl}}, \quad (2.8)$$

with

$$\alpha_{\text{cl}}(r) \sim Z_1 K_0 \left[\frac{r}{R_v} \right] + \ln \left[\frac{r}{2R_v} \right] + \gamma. \quad (2.9)$$

at large distances. Here K_0 is the modified Bessel function, R_v is the vortex line radius, $R_v = (2c_4/c_2)^{1/2}/e$, and $\gamma = 0.57721 \dots$ the Euler constant. The numerical constant Z_1 has to be determined by solving the nonlinear coupled Euler-Lagrange equations for the Abelian Higgs model; for the case $e^2 = 8c_4$, for which an exact solution is known [15], $Z_1 = 1.7079 \dots$. The corresponding asymptotic form for the scalar field can be obtained by imposing the condition (2.7) with $A_\mu^{(n)}$ given by (2.9) [5,15].

In order to compute the fermionic path integral in Eq. (2.4) we shall follow Ref. [3]. To this end we write

$$A_\mu^{(n)} = A_{\mu\text{cl}}^{(n)} + a_\mu, \quad (2.10a)$$

$$\phi^{(n)} = \phi_{\text{cl}}^{(n)} + \varphi, \quad (2.10b)$$

where $(A_{\mu\text{cl}}^{(n)}, \phi_{\text{cl}}^{(n)})$ is taken as a *fixed* background configuration carrying the topological charge. Concerning a_μ , which replaces A_μ as the integration variable for the gauge field and belongs to the zeroth topological sector, it can be written in the form

$$a_\mu = -\frac{1}{e} \epsilon_{\mu\nu} \partial_\nu \alpha + \frac{1}{e} \partial_\mu \beta. \quad (2.11)$$

We shall work in the Lorentz gauge, for which a_μ reduces to

$$a_\mu = -\frac{1}{e} \epsilon_{\mu\nu} \partial_\nu \alpha. \quad (2.12)$$

With this, the fermionic Lagrangian becomes

$$L_{\text{fermions}} = \bar{\psi} (i \not{\partial} + e A_{\text{cl}} - i \gamma_5 \not{\partial} \alpha) \psi. \quad (2.13)$$

We now decouple fermions from a_μ by performing the change of variables:

$$\psi \rightarrow \chi = \exp[\gamma_5 \alpha(x)] \psi, \quad \bar{\psi} \rightarrow \bar{\chi} = \bar{\psi} \exp[\gamma_5 \alpha(x)]. \quad (2.14)$$

After this change we have

$$Z = \sum_n \int D a_\mu D \varphi \exp(-S_{\text{Higgs}}) J[A_{\mu\text{cl}}^{(n)}, \alpha] \times \int D \bar{\chi} D \chi \exp \left[- \int d^2 \bar{\chi} \not{D}[A_{\text{cl}}^{(n)}] \chi \right]. \quad (2.15)$$

Here $J[A_{\mu\text{cl}}^{(n)}, \alpha]$ is the Fujikawa Jacobian [4] associated with the chiral change of fermionic variables (2.14):

$$D \bar{\psi} D \psi = J[A_{\mu\text{cl}}^{(n)}, \alpha] D \bar{\chi} D \chi. \quad (2.16)$$

Two comments are here in order. First, let us stress that a change of variables completely decoupling the gauge

field A_μ is possible *only* in the $n=0$ topological sector. Otherwise, one would be passing from a Dirac operator $\mathcal{D}[A^{(n)}]$ defined in the n th topological sector to $i\partial$, the Dirac operator defined in the $n=0$ sector. For this complete decoupling one then needs a topology-changing transformation, but then the usual methods to compute Jacobians (valid on compact manifolds) cannot be applied naively. In contrast, when passing from Eq. (2.4) to Eq. (2.15) we have just decoupled a_μ , a gauge field belonging to the $n=0$ sector. Then, heat-kernel or ζ -function methods can be applied without danger, to give [3]

$$\ln J[A_{\text{cl}}^{(n)}, \alpha] = -N_F \int d^2x \left[\frac{e^2}{2\pi} a_\mu^2 - \frac{e}{\pi} \alpha F_{\text{cl}} \right], \quad (2.17)$$

where

$$F_{\text{cl}} = \epsilon_{\mu\nu} \partial_\mu A_{\nu\text{cl}}. \quad (2.18)$$

Let us note at this point that this result for the Jacobian, which agrees with that obtained by Bardakci and Crescimanno [19], differs from the one implicit in Kripfganz and Ringwald [16] calculations when studying fermion-number nonconservation in the two-dimensional Abelian Higgs model. Indeed, these authors consider a complete decoupling leading to the usual result for the Jacobian in the $n=0$ topological sector:

$$J^{(0)}[A] = \exp \left[-\frac{e^2 N_F}{2\pi} \int d^2x A_\mu^{(n)2} \right]. \quad (2.19)$$

As we stated above, the use of this result (valid in the trivial sector) for computing the Jacobian in $n \neq 0$ sectors is not justified. In fact, although the fermionic determinant without zero modes in both cases differ only in an overall multiplicative (regularizing dependent) constant, we shall see that when correlation functions including fermions are considered, the correct treatment will lead to the appearance of new terms that cannot be *a priori* discarded.

The second comment concerns the fermionic determinant in the partition function (2.15), to be evaluated in a fixed background carrying topological charge. We know that the index theorem guarantees [20] the existence of $N_F|n|$ zero modes associated with the operator $\mathcal{D}[A_{\text{cl}}^{(n)}]$ [we are considering compactified space-time,

otherwise only $N_F(|n|-1)$ zero modes are square integrable]. For $n > 0$ ($n < 0$) they correspond to right-handed (left-handed) solutions $\eta_R^m(\eta_L^m)$, which in terms of light-cone variables $z = (x_0 + ix_1)/2$, $\bar{z} = (x_0 - ix_1)/2$ take the form

$$\eta_R^m = \begin{bmatrix} z^m h(z, \bar{z}) \\ 0 \end{bmatrix}, \quad (2.20)$$

$$\eta_L^m = \begin{bmatrix} 0 \\ \bar{z}^m h^{-1}(z, \bar{z}) \end{bmatrix}, \quad (2.21)$$

where $m = 0, 1, \dots, |n|-1$,

$$h(z, \bar{z}) = e^{n\alpha_{\text{cl}}(z, \bar{z})}, \quad (2.22)$$

and α_{cl} corresponds to the Nielsen-Olesen vortex solution [5] with asymptotic behavior (2.9). Then the determinant in Eq. (2.15) is zero for any $n \neq 0$ and the partition function only picks contributions from the trivial ($n=0$) sector. However, correlation functions with an adequate number of fermionic fields, as many as needed to absorb the zero-mode contribution in a given topological sector, will pick a contribution from that sector [3]. In view of this fact, we will keep all topological sectors in the partition function and just consider the fermionic determinant regularized in such a way that all zero eigenvalues are factorized through a regulator, which will be eventually made to vanish at the end of the computations. To be more precise, we shall follow Ref. [21] and define an operator $D_\epsilon[A]$:

$$D_\epsilon[A] \equiv \mathcal{D}[A] + i\epsilon I, \quad (2.23)$$

with ϵ a positive real parameter. Then, one can show [21] that if n is the number of zero modes, the limit

$$\lim_{\epsilon \rightarrow 0} \frac{\det D_\epsilon[A]}{\epsilon^n} \equiv \det' \mathcal{D}[A] \quad (2.24)$$

exists, giving an adequate expression for the product of nonzero eigenvalues, which will be called $\det' \mathcal{D}[A]$.

Inserting (2.17) in (2.15), and taking the leading order in the semiclassical expansion around the vortex configuration (2.6)–(2.9) we finally have

$$\begin{aligned} Z = \sum_n \det(i\partial + e A_{\text{cl}}^{(n)}) \exp(-S_{\text{Higgs}}[A_{\text{cl}}^{(n)}, \phi_{\text{cl}}^{(n)}]) \\ \times \int Da_\mu D\varphi \exp(-S^{\text{II}}) \exp \left\{ -N_F \int d^2x \left[\frac{e^2}{2\pi} (a_\mu[\alpha])^2 - \left[\frac{e}{\pi} \right] \alpha F_{\text{cl}}^{(n)} \right] \right\}, \end{aligned} \quad (2.25)$$

where S^{II} is given by

$$S^{\text{II}} = \int d^2x \left[-\frac{1}{4} f_{\mu\nu}^2 + \frac{1}{2} |(\partial_\mu - ie A_{\mu\text{cl}}^{(n)})\varphi|^2 + V^{\text{II}}[\phi_{\text{cl}}^{(n)}, \varphi] + \frac{e^2 a_\mu^2}{2} |\phi_{\text{cl}}^{(n)}|^2 + 2ea_\mu \text{Im}[\phi_{\text{cl}}^{(n)}(D_\mu[A_{\text{cl}}^{(n)}]\varphi)^*] \right]. \quad (2.26)$$

where

$$\mathcal{V}^{\text{II}}[\phi_{\text{cl}}^{(n)}, \varphi] = -c_2 |\varphi|^2 + 4[\text{Re}(\varphi \phi_{\text{cl}}^{(n)*})]^2 + 2|\varphi|^2 |\phi_{\text{cl}}^{(n)}|^2. \quad (2.27)$$

III. THE PARTITION FUNCTION IN THE DILUTE-GAS APPROXIMATION

It is well known that instanton contribution to the partition function can change drastically the properties of gauge theories. For example, for compact QED in 2+1 dimensions Polyakov [22] has shown that monopoles (acting as instantons in three-dimensional Euclidean space-time) change the logarithmic Coulomb interaction typical of two-dimensional space into a linear one, thus leading to confinement of external charges in the form one hopes to happen in QCD₄. An analogous phenomenon takes place in the case of an Abelian Higgs model in two dimensions when vortex contributions are taken into account in the Euclidean version of the partition function [6–9]: The careful computation of instanton effects shows in this case a symmetry restoration of the originally spontaneously broken U(1) symmetry; moreover, the analysis of the Wilson loop shows confinement of external fractional charges.

What happens if massless fermions are coupled to the Higgs-gauge system? Concerning compact QED₃ it has been recently shown by Fradkin and one of the authors [23] that the monopole contribution, responsible for confinement when fermions are absent, is wiped out if they are included. Intuitively one can think that fermions bind monopoles and antimonopoles so tightly that instead of a charged plasma they form a gas of molecules which does not exhibit Debye screening: Polyakov's scenario [22] for confinement is then ruined.

Using the techniques developed in the preceding section we shall investigate now whether the same phenomenon happens in the case of the Abelian Higgs model. This problem was already studied by Callan, Dashen, and Gross [7] and by Raby and Ukawa [8]; we shall see, however, that the method we follow allows us to go beyond the approximation used by these authors giving a more precise picture of the issues of instantons.

Let us start by noting that in Polyakov's approach [22] to the computation of instanton effects in the dilute-gas approximation one basically analyzes the partition function Z (the vacuum-to-vacuum amplitude). Now, as we have seen in the preceding section, Z picks contributions only from the $n=0$ sector due to the existence of zero modes in nontrivial topological sectors. Even if one keeps $n \neq 0$ contributions by using the zero-mode regularization described above [Eqs. (2.23) and (2.24)], one has to take the limit in which the regulator goes to zero at the end of computations; concerning the partition function or the Wilson loop, this means that only the $n=0$ contribution remains.

In his classic paper on QED₃ [22], Polyakov teaches us that even in the $n=0$ sector instantons play an important role concerning the properties of Z . Indeed, one can construct an $n=0$ configuration by superposing an equal

number of charge ± 1 instantons and anti-instantons. If we call n_a ($n_a = \pm 1$) the instanton charge (the vortex magnetic flux in the present case), one gets a neutral configuration by imposing the condition

$$\sum_a n_a = 0. \quad (3.1)$$

Although such a superposition is not a true extreme of the classical action (equations of motion are not linear) one can show in a self-consistent way that their contribution is fundamental in the dilute-gas approximation. Now even in the $n=0$ sector the approach we have developed in the preceding section for $n \neq 0$ sectors is particularly appropriate to handle Polyakov's dilute-gas approximation, which starts by considering a change of variables precisely in the form (2.10):

$$A_\mu = A_{\mu\text{cl}}^{(\text{sup})} + a_\mu, \quad (3.2)$$

with $A_{\mu\text{cl}}^{(\text{sup})}$ a superposition of N vortices with $n_a = \pm 1$ and $\sum_{a=1}^N n_a = 0$,

$$A_{\mu\text{cl}}^{(\text{sup})} = \sum_a A_{\mu\text{cl}}^{(\pm 1)}(\mathbf{x} - \mathbf{r}_a). \quad (3.3)$$

Here the flux ± 1 vortices are described by configuration $A_{\mu\text{cl}}^{(\pm 1)}$, located at points $\mathbf{r}_a$ such that $|\mathbf{r}_a - \mathbf{r}_b| \equiv r_{ab}$ satisfies

$$r_{ab} \gg R_v. \quad (3.4)$$

Condition (3.4) corresponds to a dilute-gas approximation. As we stated above, both in the monopole and the vortex cases Debye screening arises when dynamical fermions are absent, because the “charged” plasma created by the topological configurations. We intend to analyze here whether the same phenomenon happens in the presence of massless fermions.

Actual computations along the lines described above require a careful treatment of zero modes associated with invariances of the original model. Indeed, when one studies the one-loop correction to the effective action, i.e., the contribution of terms quadratic in fluctuations in (2.25), one expands (a_μ, φ) in a basis (a_μ^n, φ^n) which diagonalizes the quadratic form S^{II} :

$$a_\mu = \sum_m c_m a_\mu^m, \quad \varphi = \sum_m b_m \varphi^m, \quad (3.5)$$

so that the path-integral measure can be written in the form

$$Da_\mu D\varphi = \prod_m dc_m db_m. \quad (3.6)$$

Now, the solution to the classical equations of motion depends on parameters associated with the invariances of the classical action (the vortex position, the gauge parameter). Accordingly, there are translation- and gauge-invariance zero modes associated with each of these invariances which have to be handled using a collective coordinate approach. Basically, one eliminates these zero modes from the sums in Eq. (3.5) (so that integrals on the remaining c_m and b_m are Gaussian) trading them for collective coordinates which appear through the dependence

of the background on the invariance parameters. For example, for the translation-invariance zero modes one writes

$$A_{\mu}^{\text{sup}} = \sum_a A_{\mu\text{cl}}^{(\pm 1)}(\mathbf{x} - \mathbf{r}_a) + \sum'_m c_m a_{\mu}^m, \quad (3.7)$$

$$\phi^{\text{sup}} = \sum_a \phi_{\text{cl}}^{(\pm 1)}(\mathbf{x} - \mathbf{r}_a) + \sum'_m b_m \varphi^m, \quad (3.8)$$

where a prime indicates that zero modes are not present in the Bessel-Fourier expansion. The measure reads now

$$Da_{\mu} D\varphi = \mathcal{N} \prod_a d\mathbf{r}_a \prod'_m dc_m db_m, \quad (3.9)$$

with $\mathcal{N}$ a normalization constant. One can easily convince oneself that $\mathbf{r}_a$ are the appropriate collective coordinates associated with translation invariance by noting that

$$A_{\mu\text{cl}}(\mathbf{x} - \mathbf{r}_a) \simeq A_{\mu\text{cl}}(\mathbf{x}) + \partial_{\nu} A_{\mu\text{cl}}(\mathbf{x}) r_{a\nu}, \quad (3.10)$$

or

$$A_{\mu\text{cl}}(\mathbf{x} - \mathbf{r}_a) \simeq A_{\mu\text{cl}}(\mathbf{x}) + a_{\mu\nu}^{\text{tr}} r_{a\nu}, \quad (3.11)$$

where we denote by $a_{\mu\nu}^{\text{tr}}$ the translation-invariance zero mode, which can be easily seen to be proportional to $\partial_{\nu} A_{\mu\text{cl}}$ (in fact, the adequately normalized zero mode is $a_{\mu\nu}^{\text{tr}} = \mathcal{N}^{1/2} F_{\mu\nu\text{cl}}$).

Gauge zero modes are eliminated using the same technique (known in this case as the Faddeev-Popov method). It is convenient to work in the so-called “natural gauge,” where orthogonality between zero and nonzero modes is

ensured [22]. Once one integrates the associated collective coordinate, one ends with a measure of the form [9]

$$Da_{\mu} D\varphi = \mathcal{N} \det^{1/2} \left[\frac{1}{e^2} \nabla^2 + |\phi_{\text{cl}}^{(n)}|^2 \right] \prod_a d\mathbf{r}_a \prod'_m dc_m db_m, \quad (3.12)$$

where the determinant in (3.12) arises after simplification of the zero-mode normalization and the Faddeev-Popov determinant [9].

We can now proceed to the evaluation of the partition function up to one loop. To this end we just repeat the steps leading from Eq. (2.15) to (2.25) by keeping the contribution of neutral vortex superpositions. Introducing the expansion (3.7) and (3.8) in the Higgs action, we shall write it in the form

$$S_{\text{Higgs}}[A_{\mu}, \phi] = S_{\text{Higgs}}[A_{\mu\text{cl}}^{\text{sup}}, \phi_{\text{cl}}^{\text{sup}}] + S^{\text{II}}, \quad (3.13)$$

where S^{II} contains all terms quadratic in the fluctuations and coincides with the expression given by Eqs. (2.26) and (2.27), except that a neutral classical configuration is now acting as a background.

Concerning the evaluation of the classical action $S_{\text{Higgs}}[A_{\mu\text{cl}}^{\text{sup}}, \phi_{\text{cl}}^{\text{sup}}]$ at the neutral vortex superposition, we just follow Refs. [9] and [22]: we divide space in two regions, one including all ± 1 vortex cores and the other corresponding to the regions outside the cores. The answer is [9]

$$S_{\text{Higgs}}[A_{\mu\text{cl}}^{\text{sup}}, \phi_{\text{cl}}^{\text{sup}}] = \frac{1}{e^2 R_v^2} \left[\sum_a \epsilon \left[\frac{\lambda}{e^2} \right] + \pi Z_1^2 \sum_{a \neq b} n_a n_b K_0 \left[\frac{|\mathbf{r}_a - \mathbf{r}_b|}{R_v} \right] \right]. \quad (3.14)$$

where $\epsilon(c_4/e_2)$ is the (Euclidean) action for a charge-1 vortex [9].

Putting all this together, we have

$$Z = \sum_{N, \{n_a\}} \int Da_{\mu} D\phi \exp \left[-S_{\text{Higgs}}[A_{\mu\text{cl}}^{\text{sup}}, \phi_{\text{cl}}^{\text{sup}}] - S^{\text{II}} - N_F \int d^2x \left[\frac{e^2}{2\pi} a_{\mu} a^{\mu} + \frac{e^2}{\pi} a^{\mu} A_{\mu\text{cl}}^{\text{sup}} \right] \right] (\det \mathcal{D}[A_{\text{cl}}^{\text{sup}}])^{N_F}. \quad (3.15)$$

We now eliminate the term linear in a_{μ} from the exponential in (3.15). The answer is

$$Z = \sum_{N, \{n_a\}} \int Da_{\mu} D\phi (\det \mathcal{D}[A_{\text{cl}}^{\text{sup}}])^{N_F} \times \exp \left[-S_{\text{Higgs}}[A_{\mu\text{cl}}^{\text{sup}}, \phi_{\text{cl}}^{\text{sup}}] + \frac{e^4}{(2\pi)^2} N_F^2 \int d\mathbf{r} d\mathbf{r}' A_{\mu\text{cl}}^{\text{sup}}(\mathbf{r} - \mathbf{r}_a) S_{\mu\nu}^{-1}(\mathbf{r}, \mathbf{r}') A_{\mu\text{cl}}^{\text{sup}}(\mathbf{r}' - \mathbf{r}_b) - S^{\text{II}} \right], \quad (3.16)$$

where $S_{\mu\nu}^{-1}(\mathbf{r}, \mathbf{r}')$ is the Green's function for the block in S^{II} corresponding to the term quadratic in a_{μ} . We shall not write its explicit form, which is rather complicated looking, but we shall just present its relevant contribution, which corresponds to the region outside the vortex core. In this region we have

$$S_{\mu\nu}^{-1}(\mathbf{r}, \mathbf{r}') = -2(\nabla^2 - R_v^{-2})^{-1}(\mathbf{r}, \mathbf{r}'). \quad (3.17)$$

or

$$S_{\mu\nu}^{-1}(\mathbf{r}, \mathbf{r}') = \frac{1}{\pi} K_0 \left[\frac{|\mathbf{r} - \mathbf{r}'|}{R_v} \right]. \quad (3.18)$$

Using the explicit form (3.18) for the Green's function and that given by Eqs. (2.8) and (2.9) for the vortex superposition, we can integrate out $\mathbf{r}$ and $\mathbf{r}'$ in (3.16). The answer is

$$\begin{aligned}
Z = & \sum_{N, \{n_a\}} \int Da_\mu D\phi (\det \mathcal{D}[A_{\text{cl}}^{\text{sup}}])^{N_F} \\
& \times \exp \left[-S_{\text{Higgs}}[A_{\mu\text{cl}}^{\text{sup}}, \phi_{\text{cl}}^{\text{sup}}] \right. \\
& - \frac{e^2 R_v^2 N_F^2}{\pi} \sum_{a,b} n_a n_b \left\{ K_0 \left[\frac{r_{ab}}{R_v} \right] + \ln \left[\frac{r_{ab}}{2R_v} \right] + \gamma + Z_1(2-Z_1) \left[\frac{r_{ab}}{2R_v} K_1 \left[\frac{r_{ab}}{R_v} \right] - \frac{1}{2} \right] \right. \\
& \left. \left. + Z_1^2 \left[\frac{r_{ab}^2}{8R_v^2} K_2 \left[\frac{r_{ab}}{R_v} \right] - \frac{1}{4} \right] \right\} - S^{\text{II}} \right]. \quad (3.19)
\end{aligned}$$

We can now proceed to integrate fluctuations using the measure defined by Eq. (3.9) and evaluate explicitly the fermion determinant and the Higgs action using the superposition (3.2) in the asymptotic region. The final result is

$$Z = \sum_{N, \{n_a\}} \frac{\zeta^N}{N!} \int \prod_{a=1}^N d\mathbf{r}_a \exp \left[-\beta' \sum_{a \neq b} n_a n_b K_{ab} \right], \quad (3.20)$$

with

$$K_{ab} = \frac{\pi Z_1^2}{\beta' R_v^2 e^2} V_{ab}^{\text{Higgs}} + V_{ab}^{\text{fermions}}. \quad (3.21)$$

For reasons which will become clear later we have factorized $\beta' = N_F(N_F e^2 R_v^2 / \pi + Z_1)$ and defined

$$V_{ab}^{\text{Higgs}} = K_0 \left[\frac{r_{ab}}{R_v} \right], \quad (3.22)$$

$$\begin{aligned}
V_{ab}^{\text{fermions}} = & K_0 \left[\frac{r_{ab}}{R_v} \right] + \ln \left[\frac{r_{ab}}{2R_v} \right] + \gamma + \frac{N_F}{\beta'} \left[Z_1^2 + \frac{e^2 R_v^2 N_F}{\pi} Z_1(2-Z_1) \right] \left[\frac{r_{ab}}{2R_v} K_1 \left[\frac{r_{ab}}{R_v} \right] - \frac{1}{2} \right] \\
& + \frac{Z_1^2 N_F^2 e^2 R_v^2}{\beta' \pi} \left[\frac{1}{8} \left[\frac{r_{ab}}{R_v} \right]^2 K_2 \left[\frac{r_{ab}}{R_v} \right] - \frac{1}{4} \right]. \quad (3.23)
\end{aligned}$$

In Eq. (3.20) $\sum_{N, \{n_a\}}$ means a sum over all configurations $\{n_a\}$ leading to a neutral superposition of N vortices, each one with a $n_a = \pm 1$ flux; the $1/N!$ is included to eliminate double counting, and ζ is defined as

$$\zeta = \frac{1}{R_v^2} \Omega \exp \left[-\frac{1}{e^2 R_v^2} \epsilon \left[\frac{c_4}{e^2} \right] \right], \quad (3.24)$$

with Ω some function (containing determinants arising from the one-loop calculation) which can be in principle evaluated.

Let us now discuss the properties of the partition function (3.20) with K_{ab} given by (3.21)–(3.23). First, note that it has been cast in a form completely analogous to that obtained when fermions were absent (see Ref. [9]) except that in this last case V_{ab}^{fermions} was obviously zero. Concerning the fermion contribution, it arises in the process of decoupling quantum fluctuations a_μ from fermions (using the results of Sec. II) and from the evaluation of the fermion determinant $\det \mathcal{D}[A_{\text{cl}}^{\text{sup}}]$ at a neutral vortex superposition background.

It has been shown in Refs. [7–9] that the “charged” (magnetic) gas described by an interaction of the form V_{ab}^{Higgs} led to confinement of (fractional) electric charges when fermions were absent. That is, if one introduces heavy particles with fractional electric charge, evaluation

of the Wilson loop shows that they interact with a linear potential.

As we stated above, Callan, Dashen, and Gross [7] as well as Raby and Ukawa [8] have discussed the effect of adding massless fermions to the $d=2$ Abelian Higgs model by using statistical mechanical analogies (a Coulomb gas and an X - Y model). Our results, obtained using a different approach, confirm their scenario. Indeed, Z in Eq. (3.20) can be interpreted as the grand canonical partition function for an (overall neutral) gas of particles with “charges” $n_a = \pm 1$, i.e., the vortex magnetic flux becomes charges for the analog gas. The temperature is $T = \beta'$ for the X - Y model, and $T = 1/\beta'$ for the Coulomb gas, and the fugacity is ζ . From the form of V_{ab}^{fermions} given by Eq. (3.22) we see that a logarithmic interaction between widely separated vortices has been induced by massless fermions:

$$V_{ab}^{\text{fermions}} \sim \ln \left[\frac{r_{ab}}{R_v} \right] \quad \text{as } |r_{ab}| \rightarrow \infty, \quad (3.25)$$

$$V_{ab}^{\text{fermions}} \rightarrow 0 \quad \text{as } |r_{ab}| \rightarrow 0. \quad (3.26)$$

As advocated in Refs. [7] and [8] a sort of Coulomb gas for vortices is then established. Now, we know [24] that for such a kind of gas there is a phase transition at $T = T_c \approx 1/2$. For $T < T_c$ two oppositely signed charges

are bound together forming a dipole; for $T > T_c$ charges form a uniform plasma. Hence, in the former case [which corresponds to $N_F > N_{Fc}$ (with $N_{Fc} \simeq 2$ the value of N at which $T = T_c$)] instantons are ineffective: outside the pairs the fields are those of a noninstanton vacuum. The system behavior is essentially described by ordinary perturbation theory. This is what happens in particular in the large- N_F limit.

Dissociation of dipoles and Debye screening (with finite screening length), reappear for $T > T_c$. However, in contrast with what happened in the case $N_F = 0$, the gauge symmetry remains spontaneously broken (there is no symmetrizing effect of instantons) and hence there is no rising potential between a pair of *electric* charges [8].

It is interesting to note that the dilute-gas approximation we have performed following Polyakov's approach [22] has led us to a N_F^2 behavior for T in addition to a term with an N_F factor already found in Refs. [7] and [8]. (In fact, in Ref. [8] fermions have a charge g which is in principle different from the scalar charge e . Results can be compared just by identifying g^2/e^2 with $Z_1 N_F$.) We shall discuss in what follows the effect of this contribution on the properties of the model.

The second model that has been used to analyze the model when fermions are present relates an analog of the partition function (3.20) with that describing an X - Y model. Indeed, after bosonizing fermions, the authors in Ref. [8] arrive, within their approximation, at a connection with the continuum limit of the two-dimensional X - Y model with a classical Hamiltonian:

$$H = \int d^2x \{ \sigma [-\nabla^2 (1 - R_v \nabla^2) \sigma - h \cos \sigma] \}, \quad (3.27)$$

where h represents the external magnetic field and $\sigma(x)$ the spin angle measured with respect to that field.

In our approach the natural way to arrive to a continuum field theory is to start from the partition function (3.20) arising from the dilute-gas approximation and write

$$\begin{aligned} & \exp \left[-\beta' \sum_{a \neq b} n_a n_b K_{ab} \right] \\ &= \exp \left[\frac{\beta'}{2} \int d^2x d^2y \rho(\mathbf{x}) K(\mathbf{x} - \mathbf{y}) \rho(\mathbf{y}) \right], \end{aligned} \quad (3.28)$$

where

$$\rho(\mathbf{x}) = i\sqrt{2} \sum_a n_a \delta(\mathbf{x} - \mathbf{r}_a) \quad (3.29)$$

and $K(\mathbf{x})$ is obtained from $K_{ab} = K(\mathbf{r}_a - \mathbf{r}_b)$ just by making $\mathbf{r}_a - \mathbf{r}_b = \mathbf{x}$. Then, the partition function is easily seen to accommodate to

$$\begin{aligned} Z &= N \int D\sigma \exp \left[-\frac{\beta'}{2} \int d^2x (\sigma K^{-1} \sigma) \right] \\ &\times \sum_{N, \{n_a\}} \prod_j d\mathbf{r}_j \frac{\xi^N}{N!} \exp \left[i2\sqrt{\pi}\beta' \sum_a n_a \sigma(\mathbf{r}_a) \right], \end{aligned} \quad (3.30)$$

or

$$Z = N \int D\sigma \exp \left[-\beta \left[\int d^2x \frac{1}{2} (\sigma K^{-1} \sigma) - \frac{2\xi}{\beta} \cos(\sigma) \right] \right], \quad (3.31)$$

where

$$\beta = \frac{1}{4\pi\beta'} = \left[4\pi N_F \left[Z_1 + \frac{e^2 R_v^2}{\pi} N_F \right] \right]^{-1}. \quad (3.32)$$

Now it is easy to see that to arrive at the Hamiltonian (3.27) one has to ignore in K_{ab} : (i) the K_0 term arising from V_{ab}^{Higgs} and (ii) the term quadratic in N_F arising from a careful decoupling of massless fermions in the neutral superposition background. Indeed, one can see that it is the term linear in N_F appearing in K_{ab} the one which leads to an X - Y model (provided one approximates $Z_1 = 1$). We then conclude that in fact when one takes into account all one-loop effects in the dilute-gas approximation, one gets a model which is more complicated than the two-dimensional X - Y model. However, since the logarithmic interaction at large distances is present in the complete K_{ab} , the scenario described using the Coulomb gas and the X - Y analogies still holds. Concerning the N_F^2 term, obtained by a careful computation of the dilute-gas approximation, it does not modify the principal features described above. In particular, the critical value of β for the X - Y model (and the T_c value for the Coulomb gas) is $\beta_c = [2\pi(1 + \xi R_v^2)]^{-1}$; this corresponds in our case to

$$N_{Fc} \left[Z_1 + \frac{e^2 R_v^2}{\pi} N_{Fc} \right] = 2 + 4\xi R_v^2, \quad (3.33)$$

and hence the critical value found in Refs. [7] and [8], $Z_1 N_{Fc} \sim 2$ is not modified since $e^2 R_v^2 \ll 1$. In fact, the only regime for which the N_F^2 term becomes relevant is that corresponding to large N_F .

IV. CORRELATION FUNCTIONS IN TOPOLOGICALLY NONTRIVIAL SECTORS

In this section we shall discuss certain correlation functions which pick contributions from topologically nontrivial sectors. As instanton configurations we shall use a vortex superposition of the form (3.3) and perform a dilute instanton-anti-instanton expansion in each topological sector. In this way we analyze correlation functions, which are relevant to fermion-number-violating processes, going beyond the approximation considered in Ref. [16].

As we mentioned before, in order to get nonvanishing contributions from nontrivial sectors, one has to look for correlation functions in which the number of Fermi fields is such that zero-mode contributions to the partition function are absorbed. In other terms, consider the Bessel-Fourier expansion of Fermi fields in (2.15) written in the form

$$\chi_{R,L} = \sum_{\alpha} e_{R,L}^{\alpha} \chi_{R,L}^{\alpha} + \sum_m^{|n|-1} c_{R,L}^m \eta_{R,L}^m, \quad (4.1)$$

where $\begin{bmatrix} \chi_R^\alpha \\ \chi_L^\alpha \end{bmatrix}$ are eigenfunctions of $\mathcal{D}[A_{\text{cl}}^{(n)}]$ with nonvanishing eigenvalues (λ_α), $\eta_{R,L}^m$ are the zero modes given in Eqs. (2.20) and (2.21) and $e_{R,L}^\alpha, c_{R,L}^m$ are Grassmann coefficients (an analogous expansion holds for $\bar{\chi}$).

From the Grassmann integration point of view, the resulting fermionic determinant [and hence the partition function (2.25)] vanishes, because, for $n > 0$, there are differentials dc_R^m and $d\bar{c}_R^m$ which are not accompanied by $c_R^m, \bar{c}_R^m$ factors. (The same happens for $n < 0$.) A nonvanishing correlation function will then have to have a product of an appropriate number of χ and $\bar{\chi}$ fields which provide the lacking $c_{R,L}^m$ and $\bar{c}_{R,L}^m$ coefficients in a definite sector.

One can easily see that simple chirality-changing correlation functions of the form [3,19]

$$\begin{aligned} F_\pm(x_1^0, y_1^0, x_1^1, y_1^1, \dots, x_{N_F}^{|n|-1}, y_{N_F}^{|n|-1}) \\ = \int D\bar{\psi} D\psi \prod_{i=1}^{N_F} \prod_{\alpha=0}^{|n|-1} s_\pm^\alpha(x_i^\alpha, y_i^\alpha) \\ \times \exp \left[- \int d^2x \bar{\psi} \mathcal{D}[A^{(n)}] \psi \right], \end{aligned} \quad (4.2)$$

with

$$s_+^i(x_i^\alpha, y_i^\alpha) = \bar{\psi}_R^i(x_i^\alpha) \psi_R^i(y_i^\alpha), \quad (4.3)$$

$$s_-^i(x_i^\alpha, y_i^\alpha) = \bar{\psi}_L^i(x_i^\alpha) \psi_L^i(y_i^\alpha), \quad (4.4)$$

pick such a contribution from the n th topological sector. In fact $F_\pm^{(nN_F)}$ (when normalized) represents *minimal* (simplest nonvanishing) fermionic correlation function satisfying the criterion explained above.

They are called “minimal” in the sense that they have the minimal number of Fermi fields required to absorb the zero modes in the partition function. This means that if one computes a correlation function $F_+^{(kN_F)}$ ($F_-^{(kN_F)}$) with kN_F factors s_+ (s_-), sectors with topological charge $|n| > k$ will not contribute. In principle F_+ (F_-)

picks a contribution from all sectors with $|n| \leq k$ and one has to analyze each one of these sectors separately.

In particular for $k=1$ (i.e., a correlation function with only one factor $s_\pm^i$ per flavor) the only possible contributions come from the $n=0, \pm 1$ sector. One can easily show that the $n=0$ sector does not contribute and hence only topological sectors (with $n=\pm 1$) are important.

For simplicity we shall then consider correlation functions with just one s_+^i (or s_-^i) factor per flavor, but the treatment for a more general case can be straightforwardly extended. (In fact, in fermion-number-nonconservation analysis, where our approach finds a direct application, it is the $n=1$ sector which has been extensively studied [16,17].)

Thus, we are going to evaluate

$$\begin{aligned} F_+^{(N_F)}(x_i^{(1)}, x_i^{(2)}) = \int D\bar{\psi} D\psi \left[\prod_{i=1}^{N_F} \bar{\psi}_R^i(x_i^{(1)}) \psi_R^i(x_i^{(2)}) \right] \\ \times \exp \left[- \int d^2x \bar{\psi} \mathcal{D}[A^{(1)}] \psi \right], \end{aligned} \quad (4.5)$$

and then, eventually, integrate over $A_\mu^{(1)}$ and $\phi^{(1)}$ in order to compute the complete vacuum expectation value.

Let us write $A_\mu^{(1)}$ and $\phi^{(1)}$ in the form

$$A_\mu^{(1)} = A_{\mu\text{cl}}^{(1)} + a_\mu = \frac{1}{e} \epsilon_{\mu\nu} \partial_\nu \alpha_{\text{cl}}^{(1)} - \frac{1}{e} \epsilon_{\mu\nu} \partial_\nu \alpha, \quad (4.6a)$$

$$\phi^{(1)} = \phi_{\text{cl}}^{(1)} + \varphi, \quad (4.6b)$$

where we consider a vortex background $(A_{\mu\text{cl}}^{(1)}, \phi_{\text{cl}}^{(1)})$ with total magnetic flux $n=1$, since as we stated before, $n=0$ gives a vanishing contribution (the same happens for $n > 2$); moreover, sectors with $n < 0$ do not contribute to a “positive”-chirality-changing correlation function. As in Sec. III, we construct this background as a superposition of N vortices of charges $n_a = \pm 1$ such that $\sum_{a=1}^N n_a = 1$, located at points $\mathbf{r}_a$.

The fluctuation a_μ lives in the zeroth topological sector and so it can be completely decoupled from fermions. Following the steps detailed in Sec. II one gets

$$\begin{aligned} F_+^{(N_F)}(x_i^{(1)}, x_i^{(2)}) = \sum_{N, \{n_a\}} \exp \left[-N_F \int d^2x \left(\frac{e^2 a_\mu^2}{2\pi} - \frac{e}{\pi} \alpha F_{\text{cl}} \right) \right] \\ \times \int D\bar{\chi} D\chi \left[\prod_{i=1}^{N_F} \bar{\chi}_R^i(x_i^{(1)}) \chi_R^i(x_i^{(2)}) \right] \exp \left[- \sum_{i=1}^{N_F} [\alpha(x_i^{(1)}) + \alpha(x_i^{(2)})] \right] \exp[-\bar{\chi}(i\partial + e A_{\text{cl}}^{(1)})\chi]. \end{aligned} \quad (4.7)$$

Using the expansion (4.1), the fermionic path integral can be easily solved. The answer is

$$\begin{aligned} F_+^{(N_F)}(x_i^{(1)}, x_i^{(2)}) = \sum_{N, \{n_a\}} \det'[i\partial + e A_{\text{cl}}^{(1)}(N, \{n_a\})] \Delta(x_i^{(1)}, x_i^{(2)}) \\ \times \exp \left[- \int d^2x N_F \left(\frac{e^2 a_\mu^2}{2\pi} - \frac{e}{\pi} \alpha F_{\text{cl}} \right) - \sum_{i=1}^{N_F} [\alpha(x_i^{(1)}) + \alpha(x_i^{(2)})] \right], \end{aligned} \quad (4.8)$$

where

$$\Delta(x_i^{(1)}, x_i^{(2)}) = \exp \left[- \sum_{i=1}^{N_F} [\alpha_{\text{cl}}(x_i^{(1)}) + \alpha_{\text{cl}}(x_i^{(2)})] \right]. \quad (4.9)$$

Note that apart from $\Delta(x_i^{(1)}, x_i^{(2)})$, a new factor $\Pi(x_i^{(1)}, x_i^{(2)})$ appears in comparison with the calculation of Z in Sec. II.

$$\Pi(x_i^{(1)}, x_i^{(2)}) = \exp \left[- \sum_{i=1}^{N_F} [\alpha(x_i^{(1)}) + \alpha(x_i^{(2)})] \right]. \quad (4.10)$$

This term is disregarded in the approximation of Ref. [16].

We see from expression (4.8) that a fermion correlation

function [in this case $F_+^{N_F}(x_i^{(1)}, x_i^{(2)})$, but the generalization to $F_{\pm}^{(N_F)}(x_i^{(1)}, x_i^{(2)}, \dots, x_j^{(1)}, x_j^{(2)}, \dots)$ is evident], which was zero in the $n=0$ sector, becomes nontrivial when the flux one instanton contribution is taken into account. This result is relevant for the analysis of fermion-number-violating processes in toy models which, like the two-dimensional Abelian Higgs model, are habitually used as laboratories to treat phenomena one expects to happen in the standard model for electroweak interactions. Indeed, fermion-number-violating processes involve the analysis of Green's functions including gauge bosons, physical Higgs bosons and fermions, these last entering with F_+ -like factors. For example, a relevant Green's function of the type analyzed in Refs. [16] and [17] is

$$G(x_i^{(1)}, x_i^{(2)}; y_1, \dots, y_{n_w}; z_1, \dots, z_{n_h}) = \left\langle \prod_{i=1}^{N_F} \bar{\psi}_R^i(x_i^{(1)}) \psi_R^i(x_i^{(2)}) \prod_{j=1}^{n_w} A_{\mu}(y_j) \prod_{k=1}^{n_h} \eta(z_k) \right\rangle, \quad (4.11)$$

which involves $2N_F$ fermions, n_w gauge bosons, and n_h physical Higgs bosons η , and implies, through the Lehmann-Symanzik-Zimmermann (LSZ) reduction formula, the existence of fermion-number-violating processes. Expression (4.11) can be written as

$$G(x_i^{(1)}, x_i^{(2)}; y_1, \dots, y_{n_w}; z_1, \dots, z_{n_h}) = \frac{1}{Z_0} \int Da_{\mu} D\varphi \exp[-S_{\text{Higgs}}] F_+^{(N_F)}(x_i^{(1)}, x_i^{(2)}) \prod_{j=1}^{n_w} (a_{\mu} + A_{\mu\text{cl}}^{(1)}) \prod_{k=1}^{n_h} (\eta + \eta_{\text{cl}}^{(1)}). \quad (4.12)$$

In leading order of the semiclassical expansion around a flux one instanton, using the measure given by (3.12), we obtain

$$G(x_i^{(1)}, x_i^{(2)}; y_1, \dots, y_{n_w}; z_1, \dots, z_{n_h}) = \frac{1}{Z_0} \int \prod_j d\mathbf{r}_j \exp[-S^{\text{II}}] \exp[-S_{\text{cl}}] F_+^{(N_F)}(x_i^{(1)}, x_i^{(2)}) \prod_{j=1}^{n_w} A_{\mu\text{cl}}^{(1)}(y_j) \prod_{k=1}^{n_h} \eta_{\text{cl}}^{(1)}(z_k). \quad (4.13)$$

Now, proceeding as in Sec. IV we get

$$\begin{aligned} G(x_i^{(1)}, x_i^{(2)}; y_1, \dots, y_{n_w}; z_1, \dots, z_{n_h}) &= \frac{1}{Z_0} \exp \left\{ \frac{-e^2 R_v^2}{4\pi} \sum_{i \neq j=1}^{N_F} \sum_{\alpha \neq \alpha'=1,2} \left[\ln \left[\frac{|\mathbf{x}_i^{(\alpha)} - \mathbf{x}_j^{(\alpha')}|}{2R_v} \right] + K_0 \left[\frac{|\mathbf{x}_i^{(\alpha)} - \mathbf{x}_j^{(\alpha')}|}{R_v} \right] + \gamma \right] \right\} \\ &\times \sum_{N, \{n_a\}} \int \prod_j d\mathbf{r}_j \frac{\xi^N}{N!} \lambda^{N_F} \Delta(x_i^{(1)}, x_i^{(2)}) \exp \left[-\beta' \sum_{a \neq b} n_a n_b K_{ab} \right] \\ &\times \exp \left\{ \frac{-e^2 N_F R_v^2}{\pi} \sum_{a=1}^N n_a \sum_{i=1}^{N_F} \sum_{\alpha=1,2} \left[\ln \left[\frac{|\mathbf{r}_a - \mathbf{x}_i^{(\alpha)}|}{2R_v} \right] + \gamma - \frac{1}{2} + K_0 \left[\frac{|\mathbf{r}_a - \mathbf{x}_i^{(\alpha)}|}{R_v} \right] \right. \right. \\ &\quad \left. \left. + \frac{1}{2} \frac{|\mathbf{r}_a - \mathbf{x}_i^{(\alpha)}|}{R_v} K_1 \left[\frac{|\mathbf{r}_a - \mathbf{x}_i^{(\alpha)}|}{R_v} \right] \right] \right\} \\ &\times \prod_{j=1}^{n_w} A_{\mu\text{cl}}^{(1)}(y_j; \{\mathbf{r}_i\}) \prod_{k=1}^{n_h} \eta_{\text{cl}}^{(1)}(z_k; \{\mathbf{r}_i\}) \Bigg\}, \quad (4.14) \end{aligned}$$

where we have used

$$\det' \mathcal{D}[A_{\text{cl}}^{(1)}] = \lambda \exp \left[-\frac{e^2}{2\pi} \int d^2x [A_{\mu\text{cl}}^{(1)}(x)]^2 \right]. \quad (4.15)$$

This result for the determinant (without zero modes) in

the background $A_{\mu\text{cl}}^{(1)}$ can be easily obtained using heat-kernel or ξ -function regularization (see, for example, Ref. [25]); λ is a regularizing dependent constant which is independent of the topological charge and can be thought as an effective coupling constant of the chirality-violating

processes [19].

One should now perform the Fourier transform of this Green's function in order to extract its singularity structure and see if it contributes to the S -matrix element for a fermion-number-violating process. Let us stress in this context the differences between our approximation and that discussed in Ref. [16]:

(i) In this last reference the authors considered just one vortex configuration; thus, there is only one integration over vortex position $\mathbf{r}$ which is undone when Fourier transforming. In our approximation to the dilute instanton gas, we deal with a superposition of vortices (such that the total flux is one), and hence we end with N integrals over the collective coordinates $\mathbf{r}_a$ associated with each vortex position.

(ii) In our treatment we first decouple the gauge field fluctuations [$a_\mu = -(1/e)\epsilon_{\mu\nu}\partial_\nu\varphi$] from fermions (this can be done exactly) and then integrate over fermionic variables. In this way we get contributions of the form $\exp[-\varphi(x)]$ [see Eq. (2.14)], which in turn, after completing squares to get a quadratic form for a_μ , lead to the nonlocal factor:

$$\exp \left\{ \frac{-e^2 R_v^2}{4\pi} \sum_{i \neq j=1}^{N_F} \sum_{\alpha \neq \alpha'=1,2} \left[\log \left| \frac{|\mathbf{x}_i^{(\alpha)} - \mathbf{x}_j^{(\alpha')}|}{2R_v} \right| \right] + K_0 \left[\frac{|\mathbf{x}_i^{(\alpha)} - \mathbf{x}_j^{(\alpha')}|}{R_v} \right] + \gamma \right\} \quad (4.16)$$

appearing in (4.13).

(iii) For the reasons explained above we have also a new factor which entangles r_a 's with $\mathbf{x}_i$'s [see the last term in the exponential in (4.14)].

Because of all these new contributions, evaluation of the Fourier transform ($\tilde{G}$) from which the singularity structure of G can be analyzed becomes very complicated. In the approximation of Ref. [16] one can show (due to the locality of $\tilde{G}$) that the probability of anomalous fermion-number processes rises with energylike phase space, this being one of the reasons anomalous processes become strong at high energy. The nonlocality of expression (4.14) prevents such a direct analysis, but on the other hand may solve the problem of violation of unitarity, as signaled in Ref. [17].

We expect that in our approximation [Eq. (4.14)] one should still have large probabilities for fermion-number-violating processes. Concerning unitarity, we hope to report separately on this and related issues through a thorough analysis of the Green's function (4.14). The main purpose of the present paper was to develop an approach to treat the dilute instanton gas approximation when fermions are present, and in this sense, our main results are summarized in Eqs. (3.19) and (3.20) for the partition function, and (4.14) for correlation functions.

ACKNOWLEDGMENTS

F.S. was supported by CICBA, D.C., C.v.R., and M.T. were supported by CONICET, Argentina.

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